

# Insuring the Family or the Asset? Framing Effects on Insurance Demand

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## **Abstract**

Despite its potential to protect vulnerable households, demand for index insurance remains low. One explanation is product framing: in pastoralist settings, index insurance is typically marketed as “livestock insurance,” framing it around a specific asset rather than around household consumption. We test whether reframing insurance around household welfare increases demand, using a lab-in-the-field experiment with 287 couples in Northern Kenya. Participants exposed to a “family framing” that describes payouts as funds for household expenses allocated about 22% more of their budget to insurance (1.1 percentage points from a baseline of about 5%) than those exposed to the standard livestock framing. Because each spouse makes their own decisions, we observe both partners within the couple, and the family framing raises demand for both husbands and wives. The results suggest that how insurance is presented can affect demand.

Keywords: Inclusive Insurance, Microinsurance, Framing, Intrahousehold, Index Insurance

# 1 Introduction

Economic shocks often initially affect the assets or income of one individual, but they are costly because they lead to consumption cuts, which are often disproportionately borne by women and girls. Droughts, crop failures, and other shocks have been shown to adversely affect women’s and girls’ health, nutrition, and long-run human capital (Hoddinott and Kinsey, 2000; Akresh et al., 2011; Maccini and Yang, 2009; Anttila-Hughes and Hsiang, 2013; Dercon and Krishnan, 2000), and to erode women’s assets and earnings (Quisumbing et al., 2018; Campos et al., 2025). These gendered impacts extend beyond direct exposure: increases in women’s income, holding men’s constant, improve girls’ survival and raise educational attainment for all children, while increases in men’s income, holding women’s constant, worsen girls’ survival (Qian, 2008).

While these adverse effects on women and girls are well documented, many economic shocks initially fall on assets and income streams controlled by men. These shocks affect women indirectly by reducing household consumption or creating pressure to liquidate assets to smooth consumption. This dynamic is prevalent in settings where men are the primary asset owners or income earners. For example, in pastoralist settings, men are often responsible for the larger and more valuable animals (Kristjanson et al., 2014), and in crop-based agricultural settings, they are often responsible for cash crops whose value drives household income (Carr, 2008). Fischer and Qaim (2012) find evidence that the commercialization of bananas in Kenya led to an increase in male control of the crop, suggesting that even crops traditionally controlled by women may shift to male control as they gain value.

This paper studies whether emphasizing the potential benefits of insurance for insulating household consumption rather than protecting livestock can increase demand. We implemented a lab-in-the-field experiment in a well-suited setting for studying

these questions: Samburu County, Kenya. The experiment was conducted shortly after the baseline survey of an overlapping randomized controlled trial studying the effects of the index-based livestock insurance (IBLI) program originally designed by Chantarat et al. (2013), as well as a poverty graduation program that transferred significant assets to women.<sup>1</sup> These two interventions were combined based on the premise that drought shocks could diminish women’s gains from the graduation program, even if their businesses were not livestock-related. Early in the study, however, partners raised concerns that women might not be interested in livestock insurance because livestock are generally considered men’s assets in Northern Kenya.

To address this, we developed a novel insurance framing that tests whether emphasizing payouts’ role in preventing cuts to food consumption, school fees, and other household expenses, in addition to protecting livestock, increases demand. This hypothesis is supported by empirical evidence from the same insurance program in Southern Ethiopia, where insurance payouts were found to significantly increase food expenditures per adult equivalent in female-headed households (Timu et al., 2023).

We used a tablet-based simulation, *SimPastoralist*, in which participants invest in goats and purchase insurance over ten simulated ‘years.’ The insurance product and probabilities in the game were modeled after the IBLI program and real-world drought risks. The game was played with 287 couples randomly selected from the aforementioned randomized controlled trial.<sup>2</sup> In half of the sessions, insurance was framed traditionally: as a product designed to offset livestock losses. In the remaining sessions, we modified the framing to explain that payouts could be used either to support animals or to pay for household expenses such as food or school fees.

We find that reframing insurance to emphasize that payouts can protect house-

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<sup>1</sup>Zheng et al. (2023) describe results from the larger randomized controlled trial; the graduation program significantly increased women’s assets and earnings.

<sup>2</sup>IBLI launched in the neighboring Marsabit region in 2010 but was not available in Samburu County until late 2018, after this paper’s experiment was completed.

hold consumption increases the amount of insurance purchased. This novel ‘family framing’ increases the share of total spending allocated to insurance by approximately 22% overall, or about 1.1 percentage points from a baseline of about 5%. When estimated separately by gender, the effect is somewhat larger for women (about 1.3 percentage points) than for men (about 0.9 percentage points), though the difference is not statistically significant. The findings reported in this paper motivated a subsequent field intervention in the same region, in which the IBLI product was redesigned around women’s risk exposure and household expenses and sold in family-sized units rather than per-animal units, the results of which are reported in Arteaga et al. (2026). That study finds that the redesigned product increased demand by approximately 50% among households that had previously received an insurance subsidy. The two studies are closely linked. Our couples were drawn from the household panel of the overlapping randomized controlled trial that later hosted that intervention, and roughly half (48%) came from arms that had been offered subsidized insurance, the same population to which the reformulated product was later marketed. The sessions were run in mid-2018, after the panel’s baseline survey and before index insurance became commercially available in Samburu County, so this lab exercise served as an early test of the family framing the field product went on to implement at scale. Because that intervention was randomized at the household level, it cannot separate how husbands and wives individually responded. The experiment reported here is thus not the first evidence that household-oriented framing can raise demand; its contribution is to look inside the household, observing each partner’s decisions within the couple to test whether framing effects differ by gender and by women’s decision-making power.

The effect of reframing appears larger in households where women score lower on an empowerment index, though the difference in effects is not statistically significant. In households below the median, the family framing raises the insurance share by about 1.8 percentage points and is statistically significant, while in households above

the median the estimated effect is near zero and statistically insignificant. This pattern is consistent with our conceptual framework: in households where gender roles are more rigid, men and women operate in more separate economic spheres, making both partners more likely to narrowly frame livestock insurance as irrelevant to household consumption; the family framing bridges this gap. In more egalitarian households, the label matters less because the “separate spheres” friction is less salient.

This paper builds on the literature seeking to explain why index insurance uptake remains low despite its large potential benefits (Cai, 2016; Karlan et al., 2014; Carter et al., 2016). Existing research has explored many of these specific factors, including basis risk, uncertainty aversion, trust, social network effects, financial literacy, liquidity constraints, and disaster experience (Clarke, 2016; Carter et al., 2015; Cole et al., 2013; Jensen et al., 2015; Giné et al., 2008; Gaurav et al., 2011; Cai et al., 2015). Carter et al. (2017) and Jensen and Barrett (2017) provide specific overviews of the index insurance literature; Eling et al. (2014) and Platteau et al. (2017) review the determinants of microinsurance demand more broadly, and Dror and Eling (2021) and Eling and Yao (2024) survey the state and direction of microinsurance research. While much of this literature focuses on price, basis risk, and individual characteristics of potential subscribers, our results suggest that how insurance products are labeled and communicated can also affect demand.

The results also contribute to the literature on the impacts of narrow framing (Tversky and Kahneman, 1981; Rabin and Weizsäcker, 2009) on insurance purchase decisions (Gottlieb and Mitchell, 2020; Zheng, 2020; Dalhaus et al., 2020). Brown et al. (2008) show that most people favor a life annuity described in terms of the consumption it guarantees but reject the identical product described as an investment; Gottlieb and Mitchell (2020) extend this consumption-versus-investment logic to long-term care insurance. This paper extends the insight from the individual to the household, proposing a different type of narrow framing: rather than viewing a single

risk in isolation, household members may view their incomes and assets separately, overlooking how shocks to one partner’s livelihood affect the other. When one partner controls the insured asset and the other bears the consumption risk, the frame shapes both how the product is valued and which partner sees it as relevant. The alternative framing introduced here encourages a broader view of household finances: losses to men’s livestock can force cuts to household consumption or the liquidation of women’s assets.

The conceptual framework in this paper draws on a large body of work on intrahousehold models, and connects them to insurance decisions. While cooperative models assume Pareto optimal outcomes (Chiappori and Mazzocco, 2017), non-cooperative models allow for inefficiencies (Lundberg and Pollak, 1993; Carter and Katz, 1997). We draw on the non-cooperative framework of Browning et al. (2010) to argue that when household members make financial decisions separately, shocks to one partner’s assets may not be fully internalized by the other, reducing demand for insurance products framed around those assets. Focusing on a non-cooperative framework is supported by the work of McPeak and Doss (2006), who find evidence that decision-making in Northern Kenyan pastoralist households does not fit the predictions of cooperative models.

Finally, this paper contributes to a growing literature on inclusive insurance design. The inclusive insurance agenda (ILO Impact Insurance Facility, 2009; International Association of Insurance Supervisors, 2012; World Bank Group, 2016) emphasizes that product design and delivery must be tailored to reach underserved populations, including women, and evidence documents persistent gender gaps in insurance literacy, risk perception, and access (Borghans et al., 2009; Buvinic et al., 2009). Previous studies of IBLI in the region (Takahashi et al., 2016; Bageant and Barrett, 2017; Jensen et al., 2018) compared female- and male-headed households and found similar demand, but did not examine dynamics *within* households. Timu and

Kramer (2023) note that existing products are often designed in ways that limit their appeal to women, and Taye (2025) argues that IBLI’s design overlooks how gender shapes drought risk. Even well-designed products may not reduce poverty without complementary interventions (Flores-Contró et al., 2025). Our results suggest that linking payouts to household consumption may be a simple way of making insurance more inclusive.

The remainder of this paper is organized as follows. Section 2 presents the conceptual framework. Section 3 describes the experimental game and study design. Section 4 presents the results, and Section 5 concludes.

## **2 Conceptual Framework**

The experiment in this paper was motivated by the idea that the linkage between livestock loss and household consumption may not be front-of-mind when households decide whether to buy insurance. In the drylands of Northern Kenya, livestock are the primary productive assets for most households (Chantarat et al., 2013), and drought-induced livestock mortality can lead to catastrophic drops in household welfare. IBLI is designed to protect against this risk. However, if the product is framed narrowly around livestock asset replacement, household members may fail to connect insurance to smoothing household consumption. This disconnect is likely exacerbated by two factors: the gendered structure of the household economy and behavioral framing effects.

### **2.1 Intrahousehold Risk Sharing**

The demand for livestock insurance is likely to depend on the gendered economic structure of pastoralist households. As in many agrarian settings, pastoralist households in Northern Kenya have distinct economic responsibilities and control over

different assets. Although women play central roles in milk production, small-stock management, and household provisioning, men typically claim ownership of large livestock and exercise primary authority over livestock-related decisions, particularly whether and when animals are sold or transferred (Lesorogol et al., 2011; Njuki and Sanginga, 2013). Because men also undertake long-distance herding, women are often the ones who remain sedentary with children and smaller stock (Straight, 1997).

In most models of the household, risk to each partner’s income affects the consumption of the other. In non-cooperative and semi-cooperative models (Lundberg and Pollak, 1993; Carter and Katz, 1997; Browning et al., 2010), however, this linkage may be less obvious than in fully cooperative models, since each partner may manage their own budget and be responsible for particular decisions and expenditures. In our study context, men are the primary managers of livestock and are expected to sell livestock or otherwise contribute funds for household consumption, a dynamic that Carter and Katz (1997) describe as the “conjugal contract.” However, drought-induced livestock deaths are likely to lead men to default on this contract and reduce transfers to the household, forcing women to liquidate their own assets or reduce consumption to bridge the gap. This creates a specific friction for insurance demand. When insurance is framed around livestock, both partners may narrowly perceive its purpose. Men, who manage the herd, may view “livestock insurance” as a tool for replacing lost animals rather than for protecting household consumption, leading them to undervalue it relative to a product that explicitly covers household expenses. Women, who bear the indirect risk of drought-induced consumption shocks, may also discount “livestock insurance” as belonging to their husband’s sphere, either because they expect not to control a livestock-branded payout or because they expect it to be “stickily” labeled for herd reinvestment rather than household needs. The female-relevance channel and the shared narrow-framing channel are complementary and point the same way: the reframing should raise demand for both partners, most

strongly in households where gender roles are rigid and economic spheres are separate. The framework thus identifies the household’s separate-spheres structure, rather than the respondent’s sex, as the primary source of heterogeneity.

## **2.2 Behavioral Barriers: Narrow Framing and Labeling**

This gendered disconnect is further compounded by behavioral barriers that affect how individuals process financial products. The traditional marketing of IBLI implicitly assumes that individuals assess risks to their total household welfare. However, behavioral economics suggests that individuals often engage in “narrow framing,” evaluating specific risks and decisions in isolation rather than in conjunction with the broader portfolio of household risks (Tversky and Kahneman, 1981; Rabin and Weizsäcker, 2009). Brown et al. (2008) show how consequential the frame can be for insurance: most respondents in their survey experiment preferred an annuity framed as guaranteeing consumption but rejected the same product framed as an investment that could lose money. When insurance is marketed strictly as “livestock insurance,” individuals may likewise evaluate it solely within the domain of livestock management, comparing the cost of premiums to the value of the animals, rather than considering its role in stabilizing household consumption (e.g., food, school fees) during a shock. In our setting the asset and the consumption it ultimately protects are controlled by different household members, so the frame also shapes which partner perceives the product as relevant at all.

This tendency is reinforced by the concept of “labeling” or mental accounting, where individuals sort financial resources into distinct mental buckets based on their source or description (Thaler, 1990, 1999). Empirical evidence confirms that labeling influences spending: transfers labeled as “Winter Fuel Payments” are disproportionately spent on heating (Beatty et al., 2014), and similar effects have been documented

for SNAP benefits (Hastings and Shapiro, 2018) and in laboratory settings (Abeler and Marklein, 2017). If insurance is framed and labeled exclusively around livestock, payouts are likely mentally designated for herd reconstruction or fodder purchase.<sup>3</sup> This labeling may obscure the product’s value for both partners: men may mentally earmark payouts for restocking rather than recognizing the broader consumption-smoothing role, while women may see the product as irrelevant to their primary concerns of food security and school fees. Women’s economic behavior may also be particularly responsive to family-oriented framing: Cassar et al. (2016) find that women are more likely to compete when rewards benefit their children.

### 3 Experimental Game

This section describes a framed, incentivized field experiment studying how linking insurance to household public goods rather than male-owned assets or income affects demand. It was conducted in Samburu County, Kenya, where most household livelihoods are based on herding livestock. The region is populated primarily by members of the Samburu tribe, and the household’s livestock herd and income are traditionally controlled by men, with women earning small amounts of income by tending to small stock (goats, sheep, or chickens) at home and/or operating kiosks. As in the framework of the previous section, strongly patriarchal norms mean men consistently have larger expected incomes and greater say in livestock decisions, and the region faces frequent droughts that cause large shocks to livestock assets and hence incomes. In focus group discussions with Samburu communities, Ongoro and Ogara (2012) found that women reported food shortages and disease as consequences of droughts, while men reported livestock losses and increased conflict risk, illustrating the gendered

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<sup>3</sup>Two behavioral channels, relabeling an otherwise-fungible payout versus correcting a belief that the payout is earmarked for restocking, generate the same prediction here, and we do not attempt to separate them; the family framing does both simultaneously (see Section 5).

nature of drought impacts. Finally, the IBLI program was being rolled out in Samburu County shortly after these experimental sessions were completed. This program has traditionally linked payouts to replacing lost livestock, which according to the model above may provide limited benefit to women relative to a program that insures household public goods.

The experiment is based on SimPastoralist, a tablet-based game that simulates pastoralist life with probabilities and payoffs that approximate the real world. We collected data in 56 sessions with a total of 287 couples. In half of the sessions, insurance was framed around livestock to mirror existing IBLI products; in the other half, it was framed to associate payouts with household public goods.

Framed field experiments have been used extensively to study index insurance, both as educational tools that can increase demand (Patt et al., 2009; Cai and Song, 2017; Vasilaky et al., 2020; Janzen et al., 2021) and to study how risk preferences, loss aversion, and uncertainty aversion shape insurance decisions (Brick and Visser, 2015; McIntosh et al., 2019; Norton et al., 2014; Serfilippi et al., 2020).

### 3.1 SimPastoralist

We designed SimPastoralist in close collaboration with local Samburu enumerators, who were young people from the study area who grew up in pastoralist households. The game lasts 10 rounds, and in each round players allocate their budget between insurance, goats (investment), and education. Figure 1 shows the main screen of the SimPastoralist game (see Appendix A for additional game details).

[Figure 1 about here.]

Players start with a relatively small budget of 25,000–35,000 Kenyan Shillings.<sup>4</sup>

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<sup>4</sup>The exact starting budget (25,000, 30,000, or 35,000 KSh) was randomly assigned at the session level. In the couples version, each individual starts with half the session budget. See Appendix A for additional details.

At the beginning of each round in SimPastoralist, the player must pay 5,000 KSh for essential household consumption. If they do not have sufficient cash on hand, they can sell goats to pay these costs; if they still cannot pay, the game ends early with a final score of zero. Otherwise it ends after 10 rounds, which simulates ten years.

After paying consumption expenses, players can buy and sell goats and insurance. Goats are worth 2,000 KSh and reproduce with probability  $2/3$  in a good year and die with probability  $2/3$  in a bad year. Goats that reproduce also produce milk, generating cash income of 1,500 KSh; there is no milk production in a bad year.

Each round each player also decides whether or not to send their children to school. School costs 2,000 KSh, and six years of school are required for graduation. If the student graduates from school, the player earns a bonus of 10,000 KSh at the end of the game.<sup>5</sup>

Insurance is sold in discrete units at a premium of 350 KSh per unit, and each unit pays 1,000 KSh in the event of a drought and nothing otherwise. Because droughts occur with probability  $1/3$ , the expected payout is 333 KSh per unit and the premium carries a loading of about 5% over the actuarially fair price, an expected loss ratio of roughly 95%.<sup>6</sup> This markup is modest and probably smaller than that of the commercial index insurance available in the region, though we do not have the historical loss data needed to calculate that product’s actual markup.

The game’s risk parameters were calibrated to the region’s drought exposure. The  $1/3$  annual drought probability roughly reflects the historical frequency of severe forage deficits in Northern Kenya’s rangelands documented in the design of the IBLI product itself (Chantarat et al., 2013), and the prices and probabilities were refined

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<sup>5</sup>Nearly all participants of both genders chose to send their children to school until they graduated, meaning there is no significant variation in schooling decisions. This likely reflects the high cultural value placed on education in the region, and it means the game’s budget allocation effectively reduces to a three-way choice among goats, insurance, and cash savings.

<sup>6</sup>In coverage terms, a unit indemnifies half the value of a goat (a 1,000 KSh payout against a 2,000 KSh animal), and at 350 KSh a unit costs about 1% of a player’s opening budget. Appendix Table 9 maps these parameters to the field index-insurance product offered in the study region.

with the local enumerators noted above. This risk was highly salient to our sample: in the baseline survey conducted shortly before the sessions, 92% of the households reported losing livestock in the previous year, and roughly three-quarters had lost cattle or goats. Consistent with this, participants reported without prompting that the game reflected their lives (Appendix A).

We did not separately elicit each participant’s subjective drought expectations or risk preferences. The game’s probabilities reflect the region’s objective drought history. Because the framing is randomized at the session level and the decision environment, prices, and probabilities are identical across the two arms, any heterogeneity in participants’ risk beliefs or risk attitudes is balanced across framings; it can shift the overall level of insurance demand but does not confound the framing contrast we estimate.

After each allocation decision, the good/bad year outcome is randomly determined and animated on screen: in good years rain falls and goats reproduce and produce milk; in bad years goats die. The game reports gains and losses after each round. The final score is the sum of the value of the goats in their herd and the cash in their wallet, plus a bonus of 10,000 KSh if their child finished school. Because data collection is integrated into the game, it moves quickly, allowing participants to practice several times before the incentivized round.

In the two-player ‘couples’ version of the game, spouses play simultaneously on a split screen and can transfer resources between each other. Otherwise the game is identical: both players experience the same shocks and make individual investment and insurance decisions in response to them.

## 3.2 Sampling and Sessions

Four groups of enumerators, each assigned to a different part of Samburu County, hosted one experimental session per day over 14 days between July 26 and August 10, 2018, for a total of 56 sessions. The sampling frame was the household panel of an overlapping randomized controlled trial in Samburu County that evaluated the BOMA Rural Entrepreneur Access (REAP) graduation program crossed with an offer of index-based livestock insurance.<sup>7</sup> Sampling proceeded in three stages. First, within each participating village BOMA used a participatory wealth-ranking exercise to identify the poorest households, from which women were enrolled in the panel; the panel therefore over-samples asset-poor households by design. Second, we drew a random subset of these women into the experiment and asked each to bring her husband, yielding 287 couples. Third, sessions, framing, and starting budgets were randomly assigned by computer in advance (see below). The resulting sample spans the field trial’s arms: about half of the couples (48%) came from households that had been offered insurance and 37% from households offered the graduation program, with the remainder in control arms; 80% are in the program’s “poor” targeting tier and 20% in a more vulnerable tier. The households are a mix of fully sedentary (46%), nomadic (30%), and partially sedentary (24%) pastoralists, averaging six members.

Two features bound external validity. First, because of the poverty-targeted frame, the sample is more asset-poor and more dependent on transfers than the general Samburu population, which may make participants more liquidity-constrained and more sensitive to framing. This is, however, precisely the population that inclusive-insurance initiatives target, so the results are directly relevant for program design even if they may not generalize to wealthier pastoralists. Second, the frame is a single county and one program’s beneficiary pool. Northern Kenya is ethnically and

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<sup>7</sup>The BOMA REAP Program is a poverty graduation program similar to the well-known BRAC Program documented in Banerjee et al. (2015) and elsewhere.

linguistically diverse (Samburu, Rendille, Turkana, Borana, Somali and other pastoralist groups), and perceptions of livestock, gender roles, and insurance may differ across them. Our participants are drawn almost entirely from Samburu-speaking communities, which may limit applicability to other communities. The findings should therefore be read as evidence from a poor, BOMA-eligible, predominantly Samburu population; generalization to other groups in Northern Kenya is a question for future work. At the same time, the family/livestock framing was randomized independently of the field trial’s assignments: the framing treatment is balanced across the insurance-offer, graduation, and vulnerability arms (all  $p > 0.3$ ), so the framing effect we estimate is not confounded with the overlapping interventions.

The framings and starting budgets for each session were assigned by computer using a pseudo-random number generator in advance of fieldwork, and each team was informed on the morning of each session whether it would be hosting a ‘Livestock’ or ‘Family’ session. Appendix Table 21 confirms that baseline household characteristics are balanced across arms at the session level, the unit of randomization.

### 3.3 Experimental Sessions

Each experimental session started with a brief overview of the experiment and a skit introducing the idea of insurance. The skits were central to the experiment: in half of the sessions they introduced insurance as a product focused on livestock coverage during droughts, as has traditionally been done in the IBLI program. Each skit is oriented around two women discussing a drought affecting their families. In the ‘livestock framing’ the woman introducing insurance says:

“Have you heard about the new insurance program? It sends money to your M-Pesa account when droughts strike. We also lost half our goats and cattle, but we’ll be able to replace them using the insurance money.”

In the ‘family framing,’ which links insurance to household public goods, the woman introducing insurance instead focuses on the benefits for household welfare.

“Have you heard about the new insurance that helps families in this kind of situation? It sends money to your M-Pesa account when droughts strike to help cover household expenses. We also lost half our goats and cattle, but at least we’ll be able to pay school fees and buy food.”

See Appendix A for the full text of the scripts used in the introductory skits.

The two scripts differ along several dimensions beyond the product label. Relative to the livestock framing, the family framing (i) relabels the product as “insurance that helps families in this kind of situation”; (ii) states that the payout can be spent on household expenses, school fees, and food rather than only on replacing animals, in effect telling participants that the cash is fungible; (iii) makes household consumption salient by mentioning cut-back meals and reduced milk; and (iv) describes coverage per family member rather than per animal. The family framing does not remove the livestock framing; rather, it adds the household welfare language on top of the standard livestock description, so that participants hear about both uses for payouts (see Appendix A for full scripts). Because the family script is additive, it is also somewhat longer, so the amount of explanation participants heard is bundled with its content. This means the design identifies the combined effect of a family-oriented communication package and cannot attribute it to any single component. The game that participants play after the skit is identical across treatments: insurance in Sim-Pastoralist is sold in discrete units at 350 KSh each regardless of framing, with no per-animal or per-person pricing in the game itself. The skit therefore shapes how participants think about the product, while the product purchased and the decision environment are the same in both conditions.

By design, the game presents a simplified, high-quality insurance product: the

loading is small, payouts are automatic, and the delivery and payment hassle, limited trust, and premium-timing frictions that depress real-world take-up are absent. The insurance depicted in the game is not perfect, though: because the number of a player’s goats that die in a drought is random, the fixed per-unit payout does not exactly offset a player’s realized loss, so idiosyncratic basis risk remains. It does, however, remove the index basis risk of the field product, whose satellite-based trigger can fail to fire despite an individual’s losses. The experiment therefore does not identify demand or take-up under field conditions; holding product quality and the choice environment fixed, it isolates the effect of framing. Appendix Table 9 maps the game’s parameters to the field product offered in the study region: the commercial product carried the frictions the game strips out, and, because these households hold very little cash (median reported savings of zero), even a single unit’s premium was large relative to cash on hand.

Following the introductory skit, participants were paired with enumerators and played two practice rounds without insurance and two with insurance to learn the game mechanics. They then played the incentivized individual game followed by the couples version (again preceded by two practice rounds), knowing that one of the two incentivized games would be randomly selected for real-world payment at a rate of 1 KSh per 500 game KSh, with incentives paid individually in both games. All participants also received 500 KSh for their time.

### 3.4 Data

The sample consists of 287 couples. Of those, 151 were assigned to the family framing and 136 were assigned to the livestock framing.<sup>8</sup> Because we drew this sample from a

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<sup>8</sup>Because treatment was randomized at the session level, a small number of individual records whose framing label did not match the session’s assigned framing (likely due to data entry errors) were reassigned to the session majority framing.

larger randomized controlled trial, we had detailed information on household characteristics available. Table 1 reports household-level means by framing arm for eighteen baseline characteristics spanning demographics, herd composition, wealth and mobility, aid receipt, livestock losses, and insurance awareness. Because treatment was randomized at the session level, each difference is tested with standard errors clustered by session. The arms are balanced on all but two characteristics: family-framing households are less likely to herd cattle ( $p = 0.04$ ) and slightly less likely to be aware of the KLIP insurance program ( $p = 0.07$ ); two marginally significant differences across eighteen tests are roughly what chance produces. The estimated framing effect is essentially unchanged when the balance covariates are added as controls (0.010, versus 0.011 without them, still significant at the 5% level), so these imbalances do not drive our results. Our preferred specifications control for herd size in tropical livestock units (TLU) and household size (column 4 of all regression tables). The baseline survey did not elicit risk preferences or subjective drought expectations, so we cannot test balance on those traits; the realized-loss indicator is a backward-looking proxy.

[Table 1 about here.]

The baseline survey, which was administered only to women, also contains a set of questions pertaining to decision power; Figure 2 shows the distribution of answers among the women who participated in this experiment. Responsibilities are gendered as predicted: most women said they were fully or mostly responsible for decisions about purchasing household items, fewer said the same for buying and selling livestock, and paying for children’s medical expenses and school fees falls in between. This supports the idea that women may have more control over funds earmarked for household expenses relative to livestock.

[Figure 2 about here.]

## 4 Results

This section tests whether altering the way insurance is framed affects the quantity of insurance purchased. Appendix Table 16 summarizes game behavior. On average, participants purchased about 4 units of insurance per round (worth 1,400 KSh) and held about 12 goats. Approximately 79% of player-round observations involve some insurance purchase. Mean net worth is about 45,500 KSh, so a 1 percentage point change in the insurance budget share corresponds to approximately 455 KSh.

Overall, participants allocated approximately 22% more of their budget to insurance under the family framing than under the livestock framing. Our main specifications pool data from both the individual and couples versions of the game, but results are robust to estimating them separately (Appendix Tables 12, 13, and 14).

The first question of interest is whether the family framing increases demand for insurance on average. The basic form of the regression we use to answer this question is:

$$S_{it} = \alpha + \beta_1 FF_i + \gamma X_i + \delta_t + \phi_e + \epsilon_{it} \quad (1)$$

where  $S_{it}$  is the share of the budget allocated to insurance,  $FF_i$  is a dummy variable equal to 1 when the family framing was presented and 0 when the traditional livestock framing was presented,  $X_i$  is a vector of household-level controls,  $\delta_t$  denotes round fixed effects, and  $\phi_e$  denotes field-site fixed effects.<sup>9</sup> Round fixed effects  $\delta_t$  account for learning and strategic behavior over the game, and field-site fixed effects

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<sup>9</sup>We exclude person-round observations where the participant's net worth falls below 2,000 KSh, the price of a single goat: below this level a participant cannot hold both a goat and insurance, so the central goat-versus-insurance tradeoff is unavailable. This drops about 4% of person-round observations, and the dropout rate does not differ significantly across treatment arms ( $p = 0.71$ ). The point estimates are stable as the threshold is varied from zero to 10,000 KSh (pooled 0.008 to 0.011, low-empowerment 0.014 to 0.019; Appendix Table 23). Only in the fully unscreened sample (net worth  $> 0$ ), which retains participants who cannot afford a goat, is the pooled effect marginal ( $p = 0.066$ , versus  $p = 0.011$  at the 2,000 KSh threshold); the low-empowerment effect remains significant throughout ( $p = 0.042$  unscreened).

$\phi_e$  absorb systematic differences across the four study sites, which are geographically distinct areas that may differ in livelihoods and baseline insurance demand. The controls  $X_i$  include baseline livestock holdings (TLU) and household size. Table 2 builds from a simple bivariate regression (column 1) to the full specification (column 4). In all specifications, standard errors are clustered at the session level, the unit at which treatment assignment varies.<sup>10</sup>

Because treatment varies across only 56 sessions, we complement the session-clustered standard errors with a design-based inference battery (MacKinnon et al., 2023): a wild cluster bootstrap with 9,999 replications (Cameron et al., 2008) and randomization inference that permutes the framing assignment across sessions (Young, 2019). Appendix Table 22 reports all three: the pooled family-framing effect is significant under every method, as is the larger low-empowerment effect.

Table 2 shows the results. The family framing increases the share of insurance purchased, with statistically significant effects in specifications that include field-site fixed effects.<sup>11</sup> Among those exposed to the livestock framing, the average budget share spent on insurance is 5.1%, compared to 6.2% under the family framing, an increase of approximately 1.1 percentage points, or roughly 500 KSh per round.<sup>12</sup>

The enumerator who collected data from a participant accounts for the largest single share of variation in insurance allocation, about a quarter, far more than the

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<sup>10</sup>Appendix Tables 17 and 18 show robustness to alternative clustering (player, household, and two-way session-player), and Appendix Table 19 shows that the effect is present in both early and late rounds. Budget is endogenous to prior decisions that may be affected by treatment, so we exclude it from main specifications; results are robust to its inclusion (Appendix Table 15).

<sup>11</sup>The  $R^2$  is small in the bivariate specification (column 1) and rises to about 0.08 with round and field-site fixed effects. This is expected and does not bear on identification: a session-level randomized treatment is orthogonal to the idiosyncratic determinants of individual allocations, so it is designed to be unbiased rather than to fit the outcome. Stable individual differences account for about 43% of the outcome’s variance and within-person round-to-round variation for the remaining 57%, little of which any observable explains. The framing effect is small relative to this variance (partial  $R^2 \approx 0.006$ ) yet precisely estimated, the pattern expected of a randomized treatment.

<sup>12</sup>Since the dependent variable is a proportion bounded between zero and one with approximately 21% of observations at zero, we also estimate a fractional response model (Papke and Wooldridge, 1996). Average marginal effects from the fractional logit are nearly identical to the OLS coefficients across all specifications and subsamples.

survey enumerator, the field site, or the round (Appendix Table 20). Due to literacy issues, each household was paired with an enumerator who helped them play the game, so it is possible this is driven by variation in the way the enumerators explained the options. The family-framing effect is robust to including a fixed effect for this enumerator (column 5 of Table 2), but our main results omit it, so the effect is not an artifact of who ran the game.

[Table 2 about here.]

[Figure 3 about here.]

## 4.1 Heterogeneity by Gender and Empowerment

We next test whether the family framing operates differently for women and men by interacting the treatment with a male indicator (Table 3). The point estimates are directionally consistent with our prediction: the effect is somewhat larger for women (about 1.3 percentage points) than for men (about 0.9 percentage points). However, the interaction term is small and statistically insignificant, so we read the gender pattern as suggestive rather than conclusive. Its direction is consistent with a female-relevance channel, but the absence of a significant difference, together with the household-level pattern documented next, points primarily to a shared narrow-framing mechanism that broadens both partners’ perception of the product.<sup>13</sup>

[Table 3 about here.]

The mechanism posited in this paper suggests that the framing effect should be strongest in households where women have less control over financial decisions. To test this, we interact the framing treatment with a “High Empowerment” dummy,

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<sup>13</sup>Appendix Table 10 shows separate regressions by gender and empowerment subgroup. The results are consistent with the interaction specification.

equal to one for households where the woman’s empowerment score is above the sample median (see Appendix C for details on the empowerment index). Table 4 presents the results. The base effect of the family framing, which now captures the effect among low-empowerment households, is large and statistically significant, ranging from 1.7 to 1.8 percentage points. The interaction term is negative, indicating that the framing effect is attenuated in high-empowerment households, though the interaction term does not reach statistical significance.<sup>14</sup>

[Table 4 about here.]

The continuous empowerment index points the same way: each unit increase in empowerment is associated with a framing effect roughly 0.8 percentage points smaller, though again this interaction is not statistically significant (Appendix Table 11).

Subsample regressions confirm that the effect is driven almost entirely by households with lower female empowerment. In “Low Empowerment” households (Table 5), the family framing increases the budget share allocated to insurance by 1.7 to 1.9 percentage points, highly significant in specifications with field-site fixed effects; in “High Empowerment” households (Table 6), the effect is essentially zero and statistically insignificant.

[Table 5 about here.]

[Table 6 about here.]

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<sup>14</sup>The empowerment index could proxy for education or wealth, but the pattern is robust to controlling for both spouses’ schooling and baseline livestock holdings (TLU), and correlations with these observables are modest ( $r < 0.20$ ). Because the index is self-reported, we also test for heterogeneity on *objective* proxies for the woman’s household position (female- vs male-headed household, wife’s schooling relative to her husband’s, and a standardized composite). The family-framing effect is again significant in lower-power households under each objective measure, though, as with the self-reported index, the low-versus-high difference is not itself statistically significant (Appendix Table 24); the objective measures correlate with the index (e.g.  $r = 0.39$  with female headship), suggesting it captures genuine intra-household position rather than reporting style.

This pattern is consistent with the theoretical framework, in which the “separate spheres” friction is more salient in low-empowerment households, although, as noted, the low-versus-high difference is not statistically significant. In low-empowerment households the 1.8 percentage point increase corresponds to approximately 820 KSh per round, or roughly 2.3 additional units of insurance.

The heterogeneity that organizes the results appears to be the household’s separate-spheres structure rather than the respondent’s gender. When we split the men-only and women-only samples by empowerment, the family framing raises insurance demand for men in low-empowerment households by 1.6 percentage points ( $p < 0.05$ ) and for women in low-empowerment households by 2.2 percentage points ( $p < 0.01$ ), while the effect is statistically indistinguishable from zero for both men and women in high-empowerment households. That men in rigid-role households respond significantly, essentially as much as women, is what a shared household narrow-framing mechanism predicts, and is difficult to reconcile with a channel operating solely through women’s expected control of the payout. Because insurance in this setting is typically purchased by the male asset owner, this pattern also speaks to who is moved: the framing raises the valuation of the usual purchaser as well as less-empowered members, and men’s and women’s responses within low-empowerment households are statistically indistinguishable ( $p = 0.47$ ). The effect also holds in the couples version of the game, where partners observe each other’s choices and can transfer resources (1.2 percentage points overall,  $p < 0.05$ , Appendix Table 13; 1.9 percentage points in low-empowerment couples,  $p < 0.05$ ).

[Figure 4 about here.]

## 4.2 Extensive and Intensive Margins

Our dependent variable combines the extensive margin (whether to buy any insurance) and the intensive margin (how much to buy). Table 7 decomposes the effect. Roughly 79% of player-round observations involve some insurance purchase regardless of framing, and the extensive margin effect is not statistically significant. The framing effect operates on the intensive margin: among buyers, the family framing significantly increases purchases in the full sample (1.4 percentage points,  $p < 0.01$ ) and especially in low-empowerment households (2.3 percentage points,  $p < 0.01$ ).

[Table 7 about here.]

## 4.3 Portfolio Reallocation

Since portfolio shares must sum to one, an increase in insurance must be offset elsewhere. Table 8 estimates the treatment effect on each portfolio component. In the full sample, the only significant shift is into insurance (+1.1 percentage points). In low-empowerment households, the insurance increase of 1.8 percentage points ( $p < 0.01$ ) is offset by reductions in the cash and livestock shares, neither of which is individually significant. This suggests that the family framing draws on both precautionary cash and livestock holdings to fund additional insurance. In high-empowerment households, no component shifts significantly.

[Table 8 about here.]

## 5 Conclusion

Despite its potential to protect vulnerable households from climate shocks, demand for index insurance has remained persistently low. A large literature has examined

price, basis risk, trust, and other supply- and demand-side explanations for this puzzle (Platteau et al., 2017). This paper proposes a complementary explanation rooted in product framing: when insurance is labeled around a specific asset class, household members who do not directly control that asset may fail to recognize the product’s value for smoothing their own consumption. In pastoralist settings where men control livestock and women manage household provisioning, “livestock insurance” may be narrowly framed as a tool for herd replacement rather than a mechanism for protecting household welfare.

We test this hypothesis using a lab-in-the-field experiment with pastoralists in Northern Kenya. We find that a “family framing” that explicitly describes insurance payouts as funds for household expenses rather than asset replacement increases the share of the budget allocated to insurance by approximately 22%. The effect is concentrated in households where women have limited decision-making power, though the difference across household types is not itself statistically significant. In these households, gender roles are more rigid and men and women operate in more separate economic spheres, so the disconnect between “livestock insurance” and household consumption is greatest. In more egalitarian households, the specific label matters less because both partners already perceive the link between livestock shocks and household welfare.

Several limitations should be noted. First, the introductory skits that constitute the treatment differ along multiple dimensions: the product label, the stated use of payouts, the unit of pricing, and, because the family framing adds household-welfare language on top of the livestock description, the sheer amount of explanation. We therefore treat the manipulation as a single communication package rather than a clean label swap, and we cannot isolate which element drives the result or separate the household framing from the effect of simply providing more explanation. In particular, because payouts are cash, the framing may change how participants mentally

label an otherwise-fungible payout or may correct a mistaken belief that a livestock-branded payout is only for restocking; our instruments included no comprehension or debriefing item on payout use, and the script relabels the product and states its consumption use in the same breath, so we cannot distinguish mental labeling from learning about payout flexibility. The compound nature of the treatment is, however, also what makes it policy-relevant: an insurer or NGO redesigning its marketing would change the label, stated benefits, and narrative emphasis together, so the experiment identifies the reduced-form effect of a realistic, implementable framing change. The concentration of the effect in low-empowerment households does not by itself distinguish the household framing from the additional explanation it carries: women with less say in financial and livestock decisions may also understand the product less well at the outset, in which case more explanation of any kind would raise their demand more. We therefore read the estimate as the reduced-form effect of the communication package and do not attribute it to the household framing alone. Second, the empowerment interaction, while the subsample results are striking, does not reach statistical significance in the interaction specification, and the empowerment index may proxy for other household characteristics beyond decision-making power. Third, lab-in-the-field experiments abstract away from real-world frictions such as trust, basis risk, and liquidity constraints (Cole et al., 2013; Clarke, 2016) that could either amplify or attenuate framing effects. The simulation approach is appropriate for isolating the behavioral mechanism, but the magnitude of the effect may differ in field settings where these frictions are present.

## 5.1 Policy Implications

These limitations notwithstanding, the findings speak to the central design problem of the inclusive-insurance agenda: closing coverage and gender gaps by making a house-

hold risk-management product relevant to members who do not control the insured asset (ILO Impact Insurance Facility, 2009; International Association of Insurance Supervisors, 2012). Much of the literature on microinsurance demand has focused on price, subsidies, and financial literacy (Platteau et al., 2017; Flores-Contró et al., 2025), but our results show that how benefits are communicated, holding the product itself fixed, also moves demand, consistent with evidence that choice architecture can have large effects on behavior (Thaler and Sunstein, 2008). Reframing the same index policy around household welfare raised the insurance budget share by about 22% on average, and by roughly a third of the baseline share where women have less say over financial decisions. Because the change is a change in words rather than in price or product structure, it can be implemented at essentially no cost through agent scripts, marketing materials, and product naming. First, because the framing moves the typical purchaser as well as less-empowered members, it should raise demand through the standard sales channel even where men remain the buyers. Second, because the framing also increases demand among women, delivering the product through women’s groups or in explicitly family-sized units can reach a segment that standard livestock marketing leaves out, the combination the field reformulation of Arteaga et al. (2026) adopted.

Take-up of index insurance is low for many reasons a marketing change cannot fix: premiums are costly, basis risk erodes the value of coverage, and inexperienced households are slow to trust the product (Cole et al., 2013; Clarke, 2016). Our sample was also essentially insurance-naïve (7% had heard of the government index-insurance scheme; none had purchased a commercial policy), so the effect we measure operates before experience or subsidy has done its work. In the field experiment this study motivated, the household-oriented product raised demand among households that had previously received an insurance subsidy (Arteaga et al., 2026), consistent with subsidized experience allowing the reframing to matter. Linking payouts to household

welfare is thus a cheap, widely applicable way to make existing products more relevant to potential buyers.

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## A Experimental Details

### SimPastoralist Game Details

Figure 1 shows the main screen of the SimPastoralist game. The shield icons represent the amount of insurance the player has, and the graduation hat icon represents years of school. The balance of the wallet is shown both numerically and by adjusting the size of the visible stack of cash on the left-hand side of the screen. The enumerators explain all of these symbols to the players and update them on their balances of cash, goats, and insurance before they finalize each decision and at the beginning of each round.

The exact starting budget was randomly chosen for each session using a random number generator to be 25,000, 30,000, or 35,000 KSh. In the couples version of the game, the starting budget is divided in half, so each individual starts with a smaller

budget and the household budget is the same. The exchange rate at the time of the experiment was about 100 KSh/USD, so this corresponds to approximately 250–350 US dollars.

After the player decides how to allocate their budget between goats, insurance, and school, the clouds at the top of the screen begin moving to depict the passage of time, and the good/bad year outcome is randomly determined. In good years, rain falls on the screen and goats reproduce and produce milk. In bad years, no rain falls and goats that die fall off the screen. The game then reports the total gains/losses in animals, the amount of milk produced, and any insurance payout received.

For real-world payment, every 500 KSh in the final game score translates into 1 KSh. For example, a player ending the game with 50,000 KSh and 25 goats (a total value of 100,000 KSh) would receive a payout of 200 KSh in addition to the 500 KSh offered to all participants in compensation for their time. In both the individual and couples versions of the game, each individual has their own cash and herd of animals, so incentives were paid individually (not to couples collectively) regardless of which game was chosen for payment.

Participants reported without prompting that they felt the game was educational and reflective of their lives, and many asked to stay longer to play more after the sessions were over.

[Table 9 about here.]

## Scripts

The following are the scripts that were read by pairs of enumerators at the beginning of each session. Within each of the four teams, the same two enumerators held the roles in each session to minimize variation.

## Family Framing

**Woman 1:** It has been a tough season - no rain at all. How is the drought affecting your family?

*Kotogolo ana ng'amata meti nchan pii , aji eikunita nkolong'/lamei nkang' ino?*

**Woman 2:** We lost half our goats and cattle and have been cutting back on meals because the goats that survived aren't producing much milk. It's going to be hard to pay school fees this year, and if the children have any medical expenses we will probably have to sell even more livestock.

*Kotuata yio nkineji o nkishu, nomokure eta nkera ndaa, amaa amu meata ntare natelekunye naara kule. Kogolu abaki laata e skool e nkera, teneibisieng'u abaki nkera suom natelekunye naake kimir .*

**Woman 1:** Have you heard about the new insurance that helps families in this kind of situation? It sends money to your M-Pesa account when droughts strike to help cover household expenses. We also lost half our goats and cattle, but at least we'll be able to pay school fees and buy food.

You can also buy insurance to help offset lost animals in the event of drought. You have to pay 500 KSh per goat in August, before the rainy season begins. If the rains are poor, they send 1000 KSh for each goat to help revive your herd.

*Itining'o ana ripet ng'ejuk naret nkang'ite ta mbaa/ramat natiwenyi? Kereu ropiyiani te simu (M-pesa) nkata e nkolong' payie eyasie ramat e nkaji. Kotuata yio ntare o nkishu, keikash naa amu kindim taa atalak ropiyiani e nkera e skool, nikindim sii ainy'angu ndaa*

**Woman 2:** That sounds very helpful. How do I get this benefit?

*Panijo kotuwua keretisho kulo omon. Aji aiko payie atum ana reto?*

**Woman 1:** Well, it's not free – you have to pay 500 KSh per family member in August, before the rainy season begins. If the rains are poor, they send 1000 KSh for

each person to help pay household expenses.

If you're interested, I can introduce you to the agent who sold us our insurance.

*Maara taa pesheu, keyiari nilak ropiyiani X te ltung'ani obo le nkaji Ta lapa le esiet, eng'or ltumuren .tanaa etuesha aitibiraki nikirewakini 1000 te Ltung'ani obo le nkaji payie iasishere te ramat e nkaji. Tanaa iyieu,kaidim atirikoki ltung'ani otimira raka yio Inia ripet*

**Woman 2:** Sounds interesting - I will speak to the agent to learn more.

*Keining'o ajo keisupati kulo omon pii-kalo aiparishere ltung'ani omir ripet*

## Livestock Framing

**Woman 1:** It has been a tough season - no rain at all. How is the drought affecting your family?

*Kotogolo ana ng'amata meti nchan pii , aji eikunita nkolong'/lamei nkang' ino?*

**Woman 2:** We lost half our goats and cattle. It's going to be hard to pay school fees this year, and if the children have any medical expenses we will probably have to sell even more livestock.

*Kotuata yio nkiteeng'ata e nkishu o nkineji.kogoliki yio laata e skool tale ari, o si tinimaniki ebisiong'u nkera, kuna kuni suom naatelekunye naake kimir alakie sipitali*

**Woman 1:** Have you heard about the new insurance program? It sends money to your M-Pesa account when droughts strike. We also lost half our goats and cattle, but we'll be able to replace them using the insurance money.

*Itining'o ana ripet ng'ejuk? Kereu ropiyiani te simu (M-pesa) nkata e nkolong'. Kotuata yio ntare o nkishu, keikash naa amu kindim taa ainyang'u nkule te nenia ropiyiani e ripet.*

**Woman 2:** That sounds very helpful. How do I get this benefit?

*Panijo kotuwua keretisho kulo omon. Aji aiko payie atum ana reto?*

**Woman 1:** Well, it's not free – you have to pay 500 KSh per goat or 2500 KSh per cow in August, before the rainy season begins. If the rains are poor, they send 1000 KSh for each goat or 5000 KSh for cow to help revive your herd.

If you're interested, I can introduce you to the agent who sold us our insurance.

*Maara taa pesheu, keyiari nilak ropiyiani X te nkine nabo o ropiyiani X te nkiteng' nabo Ta lapa le esiet, eng'or ltumuren.tanaa etuesha aitibiraki nikirewakini 1000 te nkine nabo o si 5000 te nkiteng' nabo payie iramatie mboo ino. Tanaa iyieu,kaidim atirikoki ltung'ani otimiraka yio Inia ripet*

**Woman 2:** Sounds interesting - I will speak to the agent to learn more.

*Keining'o ajo keisupati kulo omon pii-kalo aiparishere ltung'ani omir ripet*

## B Additional Regression Results

[Table 10 about here.]

[Table 11 about here.]

[Table 12 about here.]

[Table 13 about here.]

[Table 14 about here.]

[Table 15 about here.]

[Table 16 about here.]

[Table 17 about here.]

[Table 18 about here.]

[Table 19 about here.]

[Table 20 about here.]

[Table 21 about here.]

[Table 22 about here.]

[Table 23 about here.]

## C Empowerment Index

The empowerment index used in this study was informed by the work of Alkire et al. (2013) on the Women’s Empowerment in Agriculture Index, but was modified to fit the livestock-based livelihoods of pastoralists. Our data collection occurred prior to the publication of the Women Empowerment in Livestock Index (Galiè et al., 2019), but does capture 5 of the 6 domains identified in that index.<sup>15</sup> This questionnaire was administered only to women (and not to men), as they are the primary participants in the larger study taking place in the region that these data were taken from.

The index we calculate is based on equal weighting of the following 10 questions, each of which is answered on a scale from 1 to 5 (1=“None – I have no say,” 2=“Little – I am rarely consulted or my opinion usually doesn’t count,” 3=“Somewhat – I make some decisions or have a say in some decisions,” 4=“Mostly – I usually make the decisions or have a major say in decisions,” 5=“Fully – I make the decisions”). We classify households where the index is above the sample median (3.3) as “High Empowerment” and those at or below the median as “Low Empowerment.”

The index can be decomposed into a household domain (questions 1–5, covering food, household items, medical expenses, school fees, and school choice) and a livestock domain (questions 6–9, covering purchasing and selling livestock). The household domain mean is higher than the livestock domain mean, consistent with women having more decision-making authority over household provisioning than over livestock transactions. The tenth question, on community-level decisions, captures a broader civic participation dimension.

The index has the following distributional properties. The mean index value is approximately 3.3 (by construction near the median), with a standard deviation of

---

<sup>15</sup>The domains are (1) decisions about agricultural production; (2) decisions related to nutrition; (3) access to and control over resources; (4) control and use of income; (5) access to and control of opportunities; and (6) workload and control over own time. This index captures information about all of those with the exception of workload and control over own time.

approximately 0.8. The interquartile range spans approximately 2.8 to 3.9, and the full range is 1.0 to 5.0. Correlations between the empowerment index and observable household characteristics are modest:  $r < 0.20$  for TLU, household size, wife’s schooling, and husband’s schooling. This suggests the index captures decision-making dynamics not fully proxied by wealth or education.

Several robustness checks support the empowerment heterogeneity results. Appendix Table 11 shows the same direction using the continuous index rather than a median split: each unit increase in empowerment is associated with a framing effect approximately 0.8 percentage points smaller, though this interaction is not statistically significant. The subsample results are robust to progressively adding controls for TLU, household size, wife’s and husband’s schooling, and wife’s and husband’s age.

The pattern does not appear to be an artifact of self-reporting. Table 24 re-estimates the heterogeneity using objective proxies for the woman’s household position in place of the self-reported index: whether the household is female- rather than male-headed, the wife’s schooling relative to her husband’s, and a standardized composite of these. Under each objective measure the family-framing effect in the lower-power group is significant (roughly 1.1 to 1.5 percentage points) and larger than in the higher-power group, where it is statistically indistinguishable from zero for two of the three measures. As with the self-reported index, however, the low-versus-high interaction is not itself statistically significant under any objective measure, so these proxies corroborate the direction of the pattern rather than establishing a significant difference. The objective measures are themselves correlated with the self-reported index ( $r = 0.39$  with female headship,  $r = 0.31$  with the composite), and the index’s interaction with framing is essentially unchanged when the objective measures are added as controls.

[Table 24 about here.]

A related concern is that the empowerment index may be correlated with other decision-relevant traits, such as risk attitudes, confidence, or baseline familiarity with insurance, so that the concentration of the effect in low-empowerment households reflects a correlate of empowerment rather than empowerment itself. This is a genuine limitation. Empowerment is not randomized. The framing treatment is, which makes the framing effect within the low-empowerment subgroup unbiased; but low- and high-empowerment households differ in any trait correlated with the index, so randomization of the framing cannot establish whether empowerment or a correlate of it accounts for the heterogeneity. What we can show is more limited. Because framing is randomized within the subgroup, the within-low-empowerment effect is robust to controls: adding a locus-of-control (agency) index, psychological distress, realized livestock loss, a self-placed wealth ladder, household-head status, and subjective livestock-decision power leaves it essentially unchanged (0.018 without these controls and 0.022 with them, both significant at the 1% level). The empowerment index is itself balanced across the framing arms ( $p = 0.23$ ).

Another question is whether it is empowerment or a correlate of it that moderates the effect. Table 25 interacts the framing treatment with each of these baseline traits in turn. The empowerment-by-framing interaction is stable in sign and magnitude, between  $-0.015$  and  $-0.019$ , and is not significant at the 5% level in any specification, so no single measured correlate accounts for the pattern. The traits the concern names most directly, risk preferences, subjective loss expectations, and baseline familiarity with the product, were not elicited, so this analysis cannot address them. We cannot isolate empowerment as the operative moderator, and, as noted in the main text, the difference between low- and high-empowerment households is not itself statistically significant; we therefore treat the empowerment pattern as suggestive.

[Table 25 about here.]

Below is a complete list of the questions. Figure 5 shows the distribution of answers to each of the questions.

- To what extent are you involved in household decisions regarding buying food for the household?
- To what extent are you involved in household decisions regarding purchasing household items (like cups)?
- To what extent are you involved in household decisions regarding paying for children's medical expenses?
- To what extent are you involved in household decisions regarding paying for children's school fees?
- To what extent are you involved in household decisions regarding which children to send to school?
- To what extent are you involved in household decisions regarding purchasing livestock for yourself?
- To what extent are you involved in household decisions regarding purchasing livestock for the whole family or household?
- To what extent are you involved in household decisions regarding selling livestock you own?
- To what extent are you involved in household decisions regarding selling livestock your household owns?
- To what extent are you involved in community-level decisions?

[Figure 5 about here.]

## List of Figures

|   |                                                                  |    |
|---|------------------------------------------------------------------|----|
| 1 | SimPastoralist . . . . .                                         | 47 |
| 2 | Distribution of responses to decision-making questions . . . . . | 48 |
| 3 | Mean Insurance Share by Treatment Arm . . . . .                  | 49 |
| 4 | Family Framing Effect by Subgroup . . . . .                      | 50 |
| 5 | Distribution of responses to decision-making questions . . . . . | 51 |

Figure 1: SimPastoralist

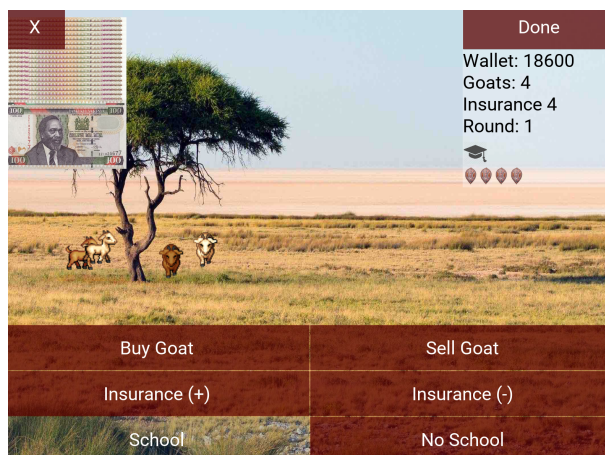
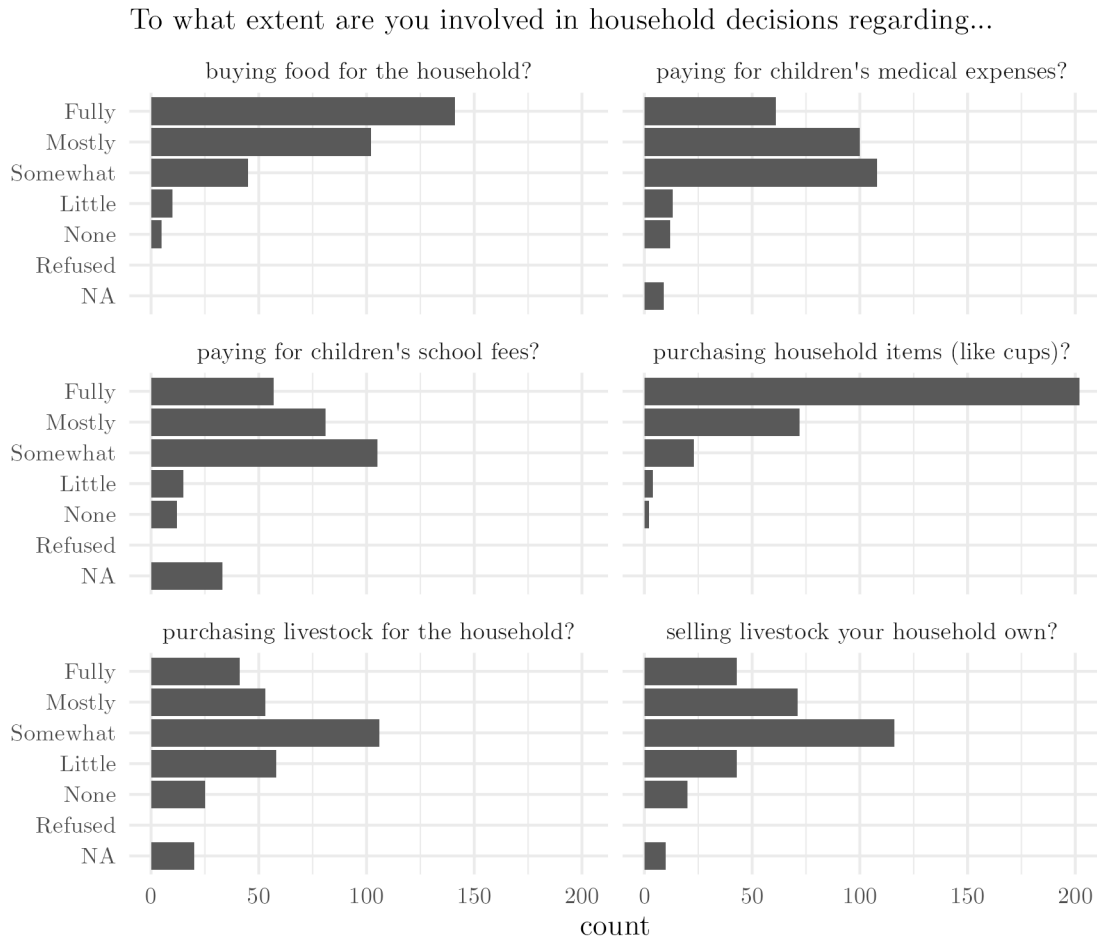
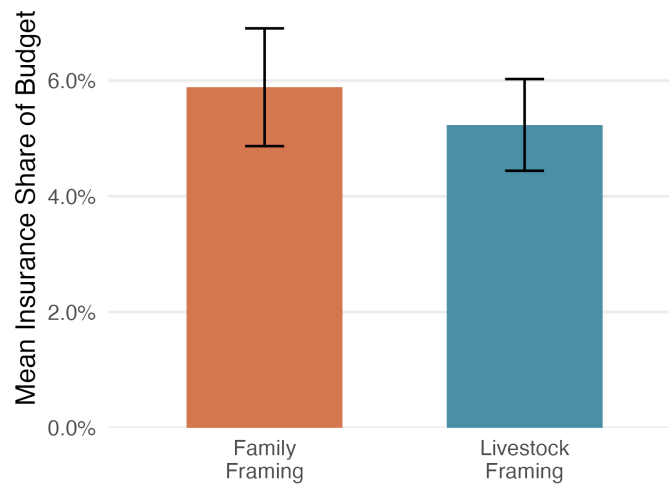


Figure 2: Distribution of responses to decision-making questions



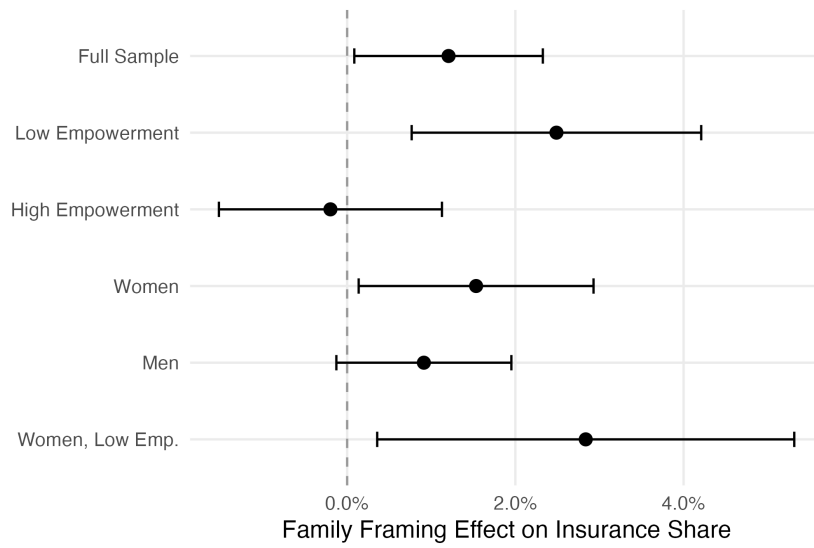
Answer text is abbreviated in the graphic above. The options in the survey read: “None – I have no say,” “Little – I am rarely consulted or my opinion usually doesn’t count,” “Somewhat – I make some decisions or have a say in some decisions,” “Mostly – I usually make the decisions or have a major say in decisions,” and “Fully – I make the decisions.”

Figure 3: Mean Insurance Share by Treatment Arm



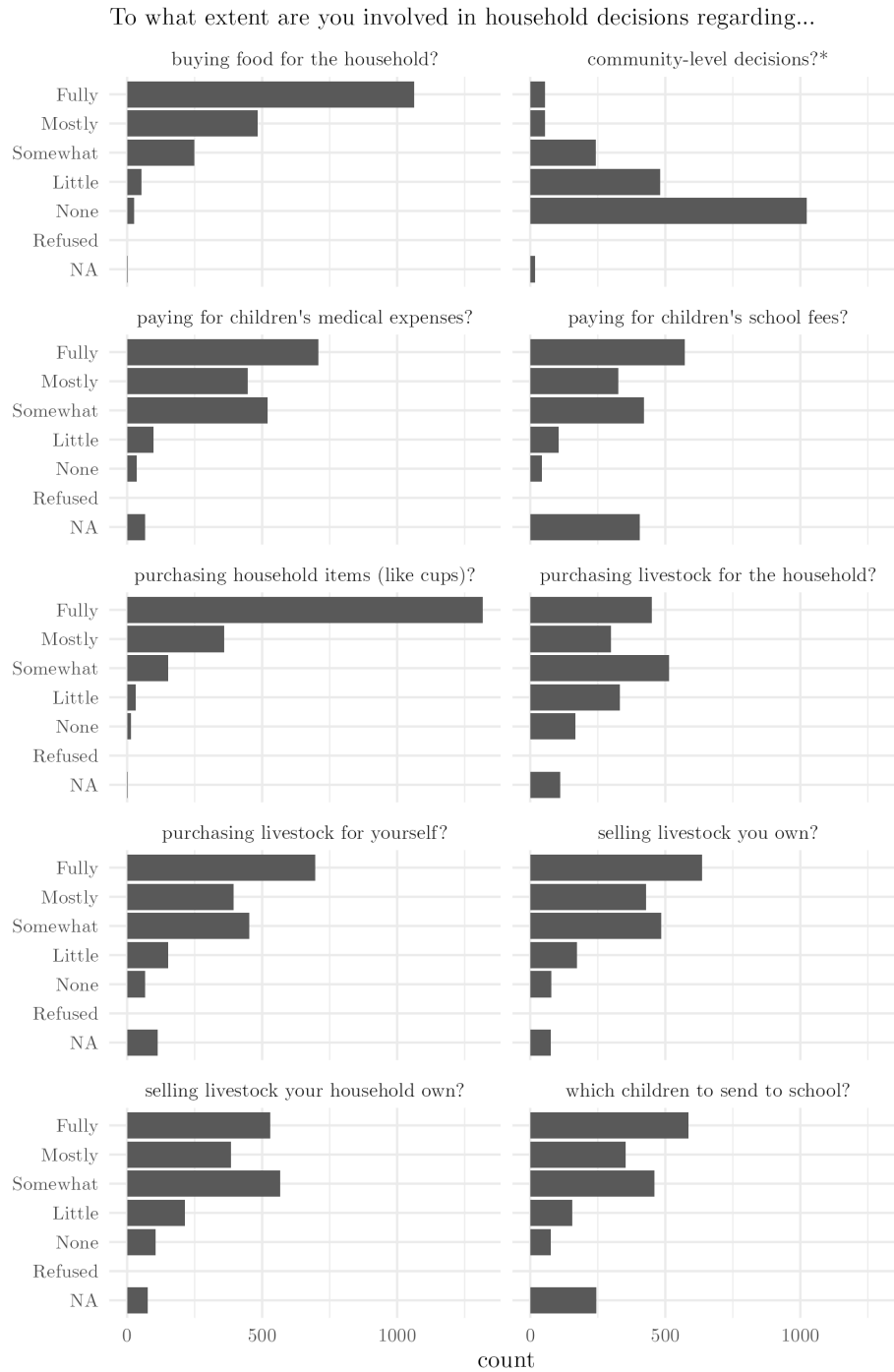
Bars show the mean share of budget allocated to insurance by treatment arm. Error bars show 95% confidence intervals computed from session-level means.

Figure 4: Family Framing Effect by Subgroup



Points show estimated coefficients on the family framing indicator from separate OLS regressions with round and field-site fixed effects and controls for TLU and household size. Horizontal bars show 95% confidence intervals based on session-clustered standard errors.

Figure 5: Distribution of responses to decision-making questions



\*This question reads "To what extent are you involved in community-level decisions?"

## List of Tables

|    |                                                                  |    |
|----|------------------------------------------------------------------|----|
| 1  | Sample Balance . . . . .                                         | 53 |
| 2  | Average Effect of Family Framing . . . . .                       | 54 |
| 3  | Gendered Effect of Family Framing . . . . .                      | 55 |
| 4  | Empowerment and the Effect of Family Framing . . . . .           | 56 |
| 5  | Effect of Family Framing: Low Empowerment Households . . . . .   | 57 |
| 6  | Effect of Family Framing: High Empowerment Households . . . . .  | 58 |
| 7  | Extensive and Intensive Margins . . . . .                        | 59 |
| 8  | Portfolio Allocation Effects of Family Framing . . . . .         | 60 |
| 9  | Game and Field Insurance Parameters . . . . .                    | 61 |
| 10 | Gendered Effect of Family Framing . . . . .                      | 62 |
| 11 | Continuous Empowerment Interaction . . . . .                     | 63 |
| 12 | Average Effect of Family Framing: Individual Game . . . . .      | 64 |
| 13 | Average Effect of Family Framing: Couples Game . . . . .         | 65 |
| 14 | Individual and Couples Results . . . . .                         | 66 |
| 15 | Average Effect of Family Framing (Budget Controls) . . . . .     | 67 |
| 16 | Descriptive Statistics: Game Behavior . . . . .                  | 68 |
| 17 | Clustering Robustness: Pooled Effect . . . . .                   | 69 |
| 18 | Clustering Robustness: Empowerment Interaction . . . . .         | 70 |
| 19 | Learning Effects: Early vs. Late Rounds . . . . .                | 71 |
| 20 | How Much Outcome Variance Each Grouping Explains . . . . .       | 72 |
| 21 | Session-Level Balance . . . . .                                  | 73 |
| 22 | Design-Based Inference for the Family-Framing Effect . . . . .   | 74 |
| 23 | Robustness to the Net-Worth Threshold . . . . .                  | 75 |
| 24 | Framing Effect by Subjective and Objective Empowerment . . . . . | 76 |
| 25 | Empowerment Moderator with Baseline-Proxy Interactions . . . . . | 77 |

Table 1: Sample Balance

| Variable                             | Livestock | Family | Difference | <i>p</i> -value | N   |
|--------------------------------------|-----------|--------|------------|-----------------|-----|
| Household size                       | 5.986     | 6.060  | 0.075      | 0.780           | 287 |
| Husband age                          | 41.935    | 42.093 | 0.157      | 0.943           | 216 |
| Husband schooling (yrs/level)        | 2.217     | 1.824  | −0.393     | 0.482           | 286 |
| Wife age                             | 31.856    | 33.227 | 1.371      | 0.349           | 273 |
| Wife schooling (yrs/level)           | 1.000     | 1.189  | 0.189      | 0.676           | 286 |
| Empowerment index                    | 3.243     | 3.359  | 0.116      | 0.361           | 287 |
| Herds camels (=1)                    | 0.123     | 0.135  | 0.012      | 0.818           | 286 |
| Herds cattle (=1)                    | 0.645     | 0.473  | −0.172     | 0.040           | 286 |
| Herds goats (=1)                     | 0.558     | 0.588  | 0.030      | 0.669           | 286 |
| Herds sheep (=1)                     | 0.841     | 0.764  | −0.077     | 0.253           | 286 |
| Herd composition score (TLU)         | 0.957     | 0.797  | −0.160     | 0.226           | 286 |
| Wealth ladder (1-4)                  | 2.225     | 2.268  | 0.044      | 0.594           | 287 |
| Fully sedentary (=1)                 | 0.420     | 0.497  | 0.076      | 0.440           | 287 |
| Receives food aid (=1)               | 0.891     | 0.919  | 0.028      | 0.505           | 287 |
| Receives other gov/NGO aid (=1)      | 0.467     | 0.480  | 0.013      | 0.858           | 285 |
| Suffered livestock loss last yr (=1) | 0.942     | 0.906  | −0.036     | 0.368           | 287 |
| Aware of KLIP (=1)                   | 0.101     | 0.040  | −0.061     | 0.065           | 287 |
| Aware of Takaful/TIA (=1)            | 0.043     | 0.020  | −0.023     | 0.224           | 287 |

Means by framing arm (one observation per household). Difference is the family-framing coefficient from a regression of each covariate on the framing indicator with standard errors clustered at the session level (the randomization unit; 56 sessions).

Table 2: Average Effect of Family Framing

|                         | (1)              | (2)              | (3)                 | (4)                | (5)                 |
|-------------------------|------------------|------------------|---------------------|--------------------|---------------------|
| Family Framing          | 0.012<br>(0.007) | 0.012<br>(0.007) | 0.011***<br>(0.004) | 0.011**<br>(0.004) | 0.010***<br>(0.003) |
| Livestock (TLU)         |                  |                  |                     | -0.004<br>(0.003)  |                     |
| Household Size          |                  |                  |                     | 0.001<br>(0.001)   |                     |
| $R^2$                   | 0.006            | 0.007            | 0.080               | 0.082              | 0.265               |
| N                       | 7923             | 7923             | 7923                | 7888               | 7923                |
| Round Effects           |                  | X                | X                   | X                  | X                   |
| Field-site Effects      |                  |                  | X                   | X                  |                     |
| Game-enumerator Effects |                  |                  |                     |                    | X                   |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.  
Standard errors clustered at the session level.

Table 3: Gendered Effect of Family Framing

|                      | (1)     | (2)     | (3)      | (4)     |
|----------------------|---------|---------|----------|---------|
| Family Framing       | 0.014*  | 0.014*  | 0.013*** | 0.012** |
|                      | (0.008) | (0.008) | (0.005)  | (0.005) |
| Man                  | 0.005   | 0.005*  | 0.006*   | 0.005*  |
|                      | (0.003) | (0.003) | (0.003)  | (0.003) |
| Man x Family Framing | -0.004  | -0.004  | -0.004   | -0.004  |
|                      | (0.005) | (0.005) | (0.005)  | (0.005) |
| Livestock (TLU)      |         |         |          | -0.004  |
|                      |         |         |          | (0.003) |
| Household Size       |         |         |          | 0.001   |
|                      |         |         |          | (0.001) |
| $R^2$                | 0.007   | 0.007   | 0.081    | 0.083   |
| N                    | 7923    | 7923    | 7923     | 7888    |
| Round Effects        |         | X       | X        | X       |
| Field-site Effects   |         |         | X        | X       |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.

Standard errors clustered at the session level.

Table 4: Empowerment and the Effect of Family Framing

|                     | (1)     | (2)     | (3)      | (4)     |
|---------------------|---------|---------|----------|---------|
| Family Framing      | 0.017*  | 0.017*  | 0.018*** | 0.018** |
|                     | (0.009) | (0.009) | (0.007)  | (0.007) |
| High Empowerment    | 0.003   | 0.003   | 0.008    | 0.008   |
|                     | (0.007) | (0.007) | (0.006)  | (0.006) |
| High Emp. x Framing | -0.011  | -0.011  | -0.016   | -0.017  |
|                     | (0.011) | (0.011) | (0.010)  | (0.011) |
| Livestock (TLU)     |         |         |          | -0.005  |
|                     |         |         |          | (0.003) |
| Household Size      |         |         |          | 0.002   |
|                     |         |         |          | (0.001) |
| $R^2$               | 0.008   | 0.008   | 0.083    | 0.085   |
| N                   | 7923    | 7923    | 7923     | 7888    |
| Round Effects       |         | X       | X        | X       |
| Field-site Effects  |         |         | X        | X       |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.

Standard errors clustered at the session level.

Table 5: Effect of Family Framing: Low Empowerment Households

|                    | (1)     | (2)     | (3)      | (4)      |
|--------------------|---------|---------|----------|----------|
| Family Framing     | 0.017*  | 0.017*  | 0.018*** | 0.019*** |
|                    | (0.009) | (0.009) | (0.007)  | (0.007)  |
| Livestock (TLU)    |         |         |          | -0.001   |
|                    |         |         |          | (0.004)  |
| Household Size     |         |         |          | 0.003*   |
|                    |         |         |          | (0.002)  |
| $R^2$              | 0.011   | 0.012   | 0.081    | 0.087    |
| N                  | 4333    | 4333    | 4333     | 4333     |
| Round Effects      |         | X       | X        | X        |
| Field-site Effects |         |         | X        | X        |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.

Standard errors clustered at the session level.

Table 6: Effect of Family Framing: High Empowerment Households

|                    | (1)              | (2)              | (3)              | (4)                 |
|--------------------|------------------|------------------|------------------|---------------------|
| Family Framing     | 0.006<br>(0.009) | 0.006<br>(0.009) | 0.002<br>(0.006) | -0.000<br>(0.005)   |
| Livestock (TLU)    |                  |                  |                  | -0.010**<br>(0.004) |
| Household Size     |                  |                  |                  | 0.001<br>(0.002)    |
| $R^2$              | 0.002            | 0.003            | 0.090            | 0.098               |
| N                  | 3590             | 3590             | 3590             | 3555                |
| Round Effects      |                  | X                | X                | X                   |
| Field-site Effects |                  |                  | X                | X                   |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.

Standard errors clustered at the session level.

Table 7: Extensive and Intensive Margins

|                    | Extensive Margin   |                   |                   | Intensive Margin    |                     |                     |
|--------------------|--------------------|-------------------|-------------------|---------------------|---------------------|---------------------|
|                    | Full Sample        | Low Emp.          | High Emp.         | Full Sample         | Low Emp.            | High Emp.           |
| Family Framing     | -0.003<br>(0.026)  | 0.021<br>(0.034)  | -0.048<br>(0.039) | 0.014***<br>(0.005) | 0.023***<br>(0.008) | 0.003<br>(0.005)    |
| Livestock (TLU)    | -0.032*<br>(0.018) | -0.033<br>(0.021) | -0.035<br>(0.027) | -0.003<br>(0.004)   | 0.003<br>(0.004)    | -0.011**<br>(0.005) |
| Household Size     | -0.002<br>(0.006)  | -0.006<br>(0.009) | 0.007<br>(0.009)  | 0.002<br>(0.001)    | 0.003*<br>(0.002)   | 0.001<br>(0.002)    |
| $R^2$              | 0.058              | 0.076             | 0.054             | 0.067               | 0.072               | 0.090               |
| N                  | 7888               | 4333              | 3555              | 6218                | 3401                | 2817                |
| Round Effects      | X                  | X                 | X                 | X                   | X                   | X                   |
| Field-site Effects | X                  | X                 | X                 | X                   | X                   | X                   |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Standard errors clustered at the session level.

Extensive margin: dependent variable is an indicator for purchasing any insurance.

Intensive margin: dependent variable is share of budget allocated to insurance, conditional on buying.

Table 8: Portfolio Allocation Effects of Family Framing

|                    | Full Sample         |                   |                   |                   | Low Empowerment     |                   |                   |                   | High Empowerment |                   |                  |                   |
|--------------------|---------------------|-------------------|-------------------|-------------------|---------------------|-------------------|-------------------|-------------------|------------------|-------------------|------------------|-------------------|
|                    | Ins.                | Lvstk.            | Cash              | Educ.             | Ins.                | Lvstk.            | Cash              | Educ.             | Ins.             | Lvstk.            | Cash             | Educ.             |
| Family Framing     | 0.011***<br>(0.004) | -0.009<br>(0.011) | -0.001<br>(0.012) | -0.002<br>(0.004) | 0.018***<br>(0.007) | -0.006<br>(0.014) | -0.008<br>(0.017) | -0.003<br>(0.004) | 0.002<br>(0.006) | -0.015<br>(0.018) | 0.015<br>(0.018) | -0.001<br>(0.005) |
| $R^2$              | 0.080               | 0.068             | 0.119             | 0.169             | 0.081               | 0.069             | 0.127             | 0.163             | 0.090            | 0.074             | 0.116            | 0.182             |
| N                  | 7923                | 7923              | 7923              | 7923              | 4333                | 4333              | 4333              | 4333              | 3590             | 3590              | 3590             | 3590              |
| Round Effects      | X                   | X                 | X                 | X                 | X                   | X                 | X                 | X                 | X                | X                 | X                | X                 |
| Field-site Effects | X                   | X                 | X                 | X                 | X                   | X                 | X                 | X                 | X                | X                 | X                | X                 |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Ins. = insurance, Lvstk. = livestock, Educ. = education. Each column estimates the same specification with a different portfolio share as the dependent variable.

Portfolio shares sum to one: Insurance + Livestock + Cash + Education = 1.

All specifications include round and field-site fixed effects.

Standard errors clustered at the session level.

Table 9: Game and Field Insurance Parameters

| Parameter            | Game (SimPas-toralist)             | Field product, Samburu 2018 <sup>a</sup>                                                      |
|----------------------|------------------------------------|-----------------------------------------------------------------------------------------------|
| Premium              | 350 KSh per unit                   | Area-mean $\approx 12\%$ of sum insured (170 KSh per shoat, 680 per family unit) <sup>b</sup> |
| Sum insured (payout) | 1,000 KSh per unit (= half a goat) | 1,400 (shoat), 5,600 (family unit), up to 19,600 (camel)                                      |
| Trigger              | Drought, probability 1/3           | Satellite forage index below a drought threshold                                              |

<sup>a</sup> Index-based livestock insurance was not sold commercially in Samburu until late 2018, after the experiment; the field figures are the terms offered to the insurance arm of the overlapping trial (division rates 10.5–13.1% of sum insured).

<sup>b</sup> In the field, premiums were subsidized through study discount coupons; the most generous covered the premium in full for up to four TLU.

Table 10: Gendered Effect of Family Framing

|                    | Full Sample         |                     | Low Empowerment     |                    | High Empowerment    |                    |
|--------------------|---------------------|---------------------|---------------------|--------------------|---------------------|--------------------|
|                    | Women               | Men                 | Women               | Men                | Women               | Men                |
| Family Framing     | 0.013***<br>(0.005) | 0.008*<br>(0.005)   | 0.022***<br>(0.008) | 0.016**<br>(0.007) | -0.001<br>(0.005)   | -0.000<br>(0.006)  |
| Livestock (TLU)    | -0.001<br>(0.004)   | -0.007**<br>(0.003) | 0.004<br>(0.005)    | -0.006<br>(0.004)  | -0.012**<br>(0.005) | -0.009*<br>(0.005) |
| Household Size     | 0.001<br>(0.001)    | 0.002<br>(0.001)    | 0.002<br>(0.002)    | 0.003*<br>(0.002)  | 0.001<br>(0.002)    | 0.001<br>(0.002)   |
| $R^2$              | 0.091               | 0.080               | 0.101               | 0.084              | 0.109               | 0.094              |
| N                  | 3968                | 3920                | 2215                | 2118               | 1753                | 1802               |
| Round Effects      | X                   | X                   | X                   | X                  | X                   | X                  |
| Field-site Effects | X                   | X                   | X                   | X                  | X                   | X                  |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.  
Standard errors clustered at the session level.

Table 11: Continuous Empowerment Interaction

|                      | (1)               | (2)               | (3)               | (4)               |
|----------------------|-------------------|-------------------|-------------------|-------------------|
| Family Framing       | 0.028<br>(0.025)  | 0.028<br>(0.025)  | 0.034<br>(0.022)  | 0.036<br>(0.023)  |
| Empowerment Index    | 0.002<br>(0.004)  | 0.002<br>(0.004)  | 0.005<br>(0.004)  | 0.005<br>(0.004)  |
| Emp. Index x Framing | -0.005<br>(0.007) | -0.005<br>(0.007) | -0.007<br>(0.006) | -0.008<br>(0.007) |
| Livestock (TLU)      |                   |                   |                   | -0.004<br>(0.003) |
| Household Size       |                   |                   |                   | 0.001<br>(0.001)  |
| $R^2$                | 0.007             | 0.007             | 0.082             | 0.084             |
| N                    | 7923              | 7923              | 7923              | 7888              |
| Round Effects        |                   | X                 | X                 | X                 |
| Field-site Effects   |                   |                   | X                 | X                 |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.

Standard errors clustered at the session level.

Table 12: Average Effect of Family Framing: Individual Game

|                    | (1)              | (2)              | (3)                | (4)                |
|--------------------|------------------|------------------|--------------------|--------------------|
| Family Framing     | 0.008<br>(0.007) | 0.008<br>(0.007) | 0.010**<br>(0.004) | 0.009**<br>(0.004) |
| Livestock (TLU)    |                  |                  |                    | -0.002<br>(0.003)  |
| Household Size     |                  |                  |                    | 0.001<br>(0.001)   |
| $R^2$              | 0.005            | 0.008            | 0.094              | 0.094              |
| N                  | 4019             | 4019             | 4019               | 4004               |
| Round Effects      |                  | X                | X                  | X                  |
| Field-site Effects |                  |                  | X                  | X                  |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.

Standard errors clustered at the session level.

Table 13: Average Effect of Family Framing: Couples Game

|                    | (1)     | (2)     | (3)     | (4)     |
|--------------------|---------|---------|---------|---------|
| Family Framing     | 0.015*  | 0.015*  | 0.012** | 0.012** |
|                    | (0.009) | (0.009) | (0.005) | (0.005) |
| Livestock (TLU)    |         |         |         | -0.006* |
|                    |         |         |         | (0.003) |
| Household Size     |         |         |         | 0.002   |
|                    |         |         |         | (0.001) |
| $R^2$              | 0.007   | 0.009   | 0.080   | 0.084   |
| N                  | 3904    | 3904    | 3904    | 3884    |
| Round Effects      |         | X       | X       | X       |
| Field-site Effects |         |         | X       | X       |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.

Standard errors clustered at the session level.

Table 14: Individual and Couples Results

|                          | (1)                 | (2)                 | (3)                 | (4)                 |
|--------------------------|---------------------|---------------------|---------------------|---------------------|
| Family Framing           | 0.008<br>(0.007)    | 0.008<br>(0.007)    | 0.009**<br>(0.004)  | 0.009**<br>(0.004)  |
| Couples                  | 0.012***<br>(0.002) | 0.012***<br>(0.002) | 0.014***<br>(0.003) | 0.013***<br>(0.002) |
| Couples x Family Framing | 0.007<br>(0.005)    | 0.007<br>(0.005)    | 0.003<br>(0.005)    | 0.004<br>(0.005)    |
| Livestock (TLU)          |                     |                     |                     | -0.004<br>(0.003)   |
| Household Size           |                     |                     |                     | 0.001<br>(0.001)    |
| $R^2$                    | 0.017               | 0.018               | 0.090               | 0.092               |
| N                        | 7923                | 7923                | 7923                | 7888                |
| Round Effects            |                     | X                   | X                   | X                   |
| Field-site Effects       |                     |                     | X                   | X                   |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.  
Standard errors clustered at the session level.

Table 15: Average Effect of Family Framing (Budget Controls)

|                    | (1)              | (2)              | (3)                 | (4)                | (5)                  | (6)                  | (7)                 | (8)                 |
|--------------------|------------------|------------------|---------------------|--------------------|----------------------|----------------------|---------------------|---------------------|
| Family Framing     | 0.012<br>(0.007) | 0.012<br>(0.007) | 0.011***<br>(0.004) | 0.011**<br>(0.004) | 0.011<br>(0.007)     | 0.011<br>(0.007)     | 0.010**<br>(0.004)  | 0.010**<br>(0.004)  |
| Budget             |                  |                  |                     |                    | -0.007***<br>(0.002) | -0.007***<br>(0.003) | -0.006**<br>(0.003) | -0.006**<br>(0.003) |
| Livestock (TLU)    |                  |                  |                     | -0.004<br>(0.003)  |                      |                      |                     | -0.005<br>(0.003)   |
| Household Size     |                  |                  |                     | 0.001<br>(0.001)   |                      |                      |                     | 0.001<br>(0.001)    |
| $R^2$              | 0.006            | 0.007            | 0.080               | 0.082              | 0.019                | 0.021                | 0.091               | 0.093               |
| N                  | 7923             | 7923             | 7923                | 7888               | 7923                 | 7923                 | 7923                | 7888                |
| Round Effects      |                  | X                | X                   | X                  |                      | X                    | X                   | X                   |
| Field-site Effects |                  |                  | X                   | X                  |                      |                      | X                   | X                   |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.

Standard errors clustered at the session level.

Table 16: Descriptive Statistics: Game Behavior

| Variable                  | Mean     | SD        | N    |
|---------------------------|----------|-----------|------|
| <i>Full Sample</i>        |          |           |      |
| Insurance units purchased | 3.99     | 7.49      | 7923 |
| Insurance share of budget | 0.057    | 0.077     | 7923 |
| Goats held                | 11.86    | 41.93     | 7923 |
| Cash held (KSh)           | 19160.87 | 52685.31  | 7923 |
| Sent child to school (%)  | 61.15    | 48.74     | 7923 |
| Net worth (KSh)           | 45501.61 | 131087.85 | 7923 |
| <i>Women</i>              |          |           |      |
| Insurance units purchased | 4.00     | 7.19      | 3988 |
| Insurance share of budget | 0.055    | 0.077     | 3988 |
| Goats held                | 12.93    | 51.42     | 3988 |
| Cash held (KSh)           | 21021.34 | 61212.31  | 3988 |
| Sent child to school (%)  | 60.31    | 48.93     | 3988 |
| Net worth (KSh)           | 49494.45 | 158173.69 | 3988 |
| <i>Men</i>                |          |           |      |
| Insurance units purchased | 3.98     | 7.78      | 3935 |
| Insurance share of budget | 0.059    | 0.076     | 3935 |
| Goats held                | 10.77    | 29.30     | 3935 |
| Cash held (KSh)           | 17275.34 | 42250.37  | 3935 |
| Sent child to school (%)  | 62.01    | 48.54     | 3935 |
| Net worth (KSh)           | 41454.99 | 95996.91  | 3935 |
| <i>Livestock Framing</i>  |          |           |      |
| Insurance units purchased | 4.27     | 9.03      | 3897 |
| Insurance share of budget | 0.051    | 0.068     | 3897 |
| Goats held                | 14.46    | 55.67     | 3897 |
| Cash held (KSh)           | 21929.14 | 64094.38  | 3897 |
| Sent child to school (%)  | 62.69    | 48.37     | 3897 |
| Net worth (KSh)           | 53604.99 | 170307.66 | 3897 |
| <i>Family Framing</i>     |          |           |      |
| Insurance units purchased | 3.72     | 5.60      | 4026 |
| Insurance share of budget | 0.063    | 0.083     | 4026 |
| Goats held                | 9.34     | 21.15     | 4026 |
| Cash held (KSh)           | 16481.30 | 38369.05  | 4026 |
| Sent child to school (%)  | 59.66    | 49.06     | 4026 |
| Net worth (KSh)           | 37657.87 | 74975.68  | 4026 |

Notes: Statistics computed over player-round observations after dropping observations with net worth below 2,000 KSh (5.0% of all observations). Insurance units are the number of livestock insured in a given round. Insurance share is the value of insurance purchased divided by net worth. Goats held is the number of goats in the player's herd at the start of the round.

Table 17: Clustering Robustness: Pooled Effect

|                    | Session            | Player              | Household          | Session+Player     |
|--------------------|--------------------|---------------------|--------------------|--------------------|
| Family Framing     | 0.011**<br>(0.004) | 0.011***<br>(0.004) | 0.011**<br>(0.005) | 0.011**<br>(0.004) |
| Livestock (TLU)    | -0.004<br>(0.003)  | -0.004<br>(0.003)   | -0.004<br>(0.003)  | -0.004<br>(0.003)  |
| Household Size     | 0.001<br>(0.001)   | 0.001<br>(0.001)    | 0.001<br>(0.001)   | 0.001<br>(0.001)   |
| $R^2$              | 0.082              | 0.082               | 0.082              | 0.082              |
| N                  | 7888               | 7888                | 7888               | 7888               |
| Round Effects      | X                  | X                   | X                  | X                  |
| Field-site Effects | X                  | X                   | X                  | X                  |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.  
All specifications include round and field-site fixed effects  
and controls for TLU and household size.  
Columns differ only in the level of clustering for standard  
errors.

Table 18: Clustering Robustness: Empowerment Interaction

|                     | Session            | Player              | Household          | Session+Player     |
|---------------------|--------------------|---------------------|--------------------|--------------------|
| Family Framing      | 0.018**<br>(0.007) | 0.018***<br>(0.006) | 0.018**<br>(0.007) | 0.018**<br>(0.007) |
| High Empowerment    | 0.008<br>(0.006)   | 0.008<br>(0.005)    | 0.008<br>(0.007)   | 0.008<br>(0.006)   |
| High Emp. x Framing | -0.017<br>(0.011)  | -0.017**<br>(0.008) | -0.017*<br>(0.010) | -0.017<br>(0.011)  |
| Livestock (TLU)     | -0.005<br>(0.003)  | -0.005*<br>(0.003)  | -0.005<br>(0.003)  | -0.005<br>(0.003)  |
| Household Size      | 0.002<br>(0.001)   | 0.002*<br>(0.001)   | 0.002<br>(0.001)   | 0.002<br>(0.001)   |
| $R^2$               | 0.085              | 0.085               | 0.085              | 0.085              |
| N                   | 7888               | 7888                | 7888               | 7888               |
| Round Effects       | X                  | X                   | X                  | X                  |
| Field-site Effects  | X                  | X                   | X                  | X                  |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.

All specifications include round and field-site fixed effects and controls for TLU and household size.

Columns differ only in the level of clustering for standard errors.

Table 19: Learning Effects: Early vs. Late Rounds

|                    | Full Sample       |                     | Low Empowerment     |                    | High Empowerment    |                    |
|--------------------|-------------------|---------------------|---------------------|--------------------|---------------------|--------------------|
|                    | Early             | Late                | Early               | Late               | Early               | Late               |
| Family Framing     | 0.006<br>(0.004)  | 0.019***<br>(0.006) | 0.016***<br>(0.006) | 0.024**<br>(0.010) | -0.007<br>(0.005)   | 0.010<br>(0.006)   |
| Livestock (TLU)    | -0.003<br>(0.003) | -0.006*<br>(0.004)  | 0.000<br>(0.003)    | -0.003<br>(0.005)  | -0.009**<br>(0.004) | -0.012*<br>(0.006) |
| Household Size     | 0.000<br>(0.001)  | 0.003**<br>(0.001)  | 0.002<br>(0.001)    | 0.004**<br>(0.002) | -0.000<br>(0.002)   | 0.002<br>(0.002)   |
| $R^2$              | 0.086             | 0.087               | 0.096               | 0.083              | 0.096               | 0.119              |
| N                  | 4997              | 2891                | 2746                | 1587               | 2251                | 1304               |
| Round Effects      | X                 | X                   | X                   | X                  | X                   | X                  |
| Field-site Effects | X                 | X                   | X                   | X                  | X                   | X                  |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Dependent variable: share of budget allocated to insurance.

Early rounds: 0–4. Late rounds: 5–10.

All specifications include round and field-site fixed effects and controls for TLU and household size.

Standard errors clustered at the session level.

Table 20: How Much Outcome Variance Each Grouping Explains

| Grouping                     | Levels | $R^2$ of insurance share |
|------------------------------|--------|--------------------------|
| Game enumerator              | 29     | 0.260                    |
| Session (date $\times$ site) | 56     | 0.118                    |
| Baseline-survey enumerator   | 29     | 0.081                    |
| Field site                   | 4      | 0.074                    |
| Round                        | 10     | 0.001                    |

Each row is the  $R^2$  of a model of the insurance share on that grouping's fixed effects alone (the between-group share of total outcome variance). The game enumerator (paired with each household to help them play the game) explains far more variance than the baseline-survey enumerator, the field site, or the round.

Table 21: Session-Level Balance

| Variable              | Livestock | Family | p-value |
|-----------------------|-----------|--------|---------|
| Household Size        | 6.050     | 6.090  | 0.905   |
| Husband Age           | 41.643    | 41.804 | 0.940   |
| Husband School        | 2.139     | 1.948  | 0.744   |
| Empowerment Index     | 3.229     | 3.363  | 0.296   |
| Wife Age              | 31.603    | 32.822 | 0.400   |
| Wife School           | 0.993     | 1.109  | 0.795   |
| Total Livestock Units | 0.969     | 0.793  | 0.172   |

Means of baseline covariates collapsed to the session level (26 Livestock, 30 Family sessions), the unit at which framing was randomized. p-values from two-sample t-tests across sessions. Joint F-test of framing on all covariates:  $F(7, 47) = 0.69$ ,  $p = 0.676$ .

Table 22: Design-Based Inference for the Family-Framing Effect

| Sample           | Coefficient<br>(session-clustered) | CR1<br>$p$ -value | Wild cluster<br>bootstrap $p$ | Randomization<br>inference $p$ |
|------------------|------------------------------------|-------------------|-------------------------------|--------------------------------|
| Pooled           | 0.0111                             | 0.009             | 0.012                         | 0.020                          |
| Low empowerment  | 0.0181                             | 0.010             | 0.015                         | 0.018                          |
| High empowerment | 0.0020                             | 0.742             | 0.766                         | 0.780                          |

Dependent variable: share of budget allocated to insurance. All models include round and field-site fixed effects; standard errors are clustered at the session level (the randomization unit; 56 sessions). CR1 is the cluster-robust  $p$ -value. The wild cluster bootstrap imposes the null (Rademacher weights,  $B = 9,999$ ). Randomization inference permutes the framing assignment across sessions ( $B = 5,000$ ). The pooled and low-empowerment effects are significant under every method, and all three methods agree that the high-empowerment effect is indistinguishable from zero.

Table 23: Robustness to the Net-Worth Threshold

| Net-worth threshold (KSh) | Pooled coef. | Pooled $p$ | Low-emp. coef. | Low-emp. $p$ | % kept |
|---------------------------|--------------|------------|----------------|--------------|--------|
| > 0                       | 0.009        | 0.066      | 0.017          | 0.042        | 100.0  |
| > 2,000                   | 0.011        | 0.011      | 0.019          | 0.008        | 96.4   |
| > 4,000                   | 0.010        | 0.009      | 0.017          | 0.009        | 91.7   |
| > 6,000                   | 0.009        | 0.022      | 0.014          | 0.021        | 87.2   |
| > 8,000                   | 0.008        | 0.024      | 0.014          | 0.019        | 82.8   |
| > 10,000                  | 0.008        | 0.022      | 0.014          | 0.016        | 76.8   |

Family-framing coefficient on the insurance share as the net-worth threshold is varied, with round and field-site fixed effects and session-clustered standard errors. “% kept” is the share of person-round observations retained. The paper’s specifications retain the 2,000 KSh threshold, below which a participant cannot afford a single goat and faces no allocation decision. The point estimates are stable across thresholds.

Table 24: Framing Effect by Subjective and Objective Empowerment

| Empowerment measure                              | Low<br>empowerment  | High<br>empowerment | Interaction<br>(High×Framing) |
|--------------------------------------------------|---------------------|---------------------|-------------------------------|
| Subjective decision index                        | 0.019***<br>(0.007) | 0.000<br>(0.005)    | −0.017<br>(0.011)             |
| Objective: female- vs male-headed household      | 0.011**<br>(0.005)  | 0.007<br>(0.014)    | 0.001<br>(0.019)              |
| Objective: wife vs husband schooling             | 0.014*<br>(0.008)   | 0.009*<br>(0.005)   | 0.001<br>(0.011)              |
| Objective: standardized composite (median split) | 0.015***<br>(0.005) | 0.003<br>(0.007)    | −0.010<br>(0.010)             |

Each row splits the sample by a different measure of the woman’s household position: the self-reported decision index, or one of three objective proxies used *in its place* (female- vs male-headed household; wife’s schooling relative to her husband’s; a standardized composite of the two and the spousal age gap). Within each split, the “Low”/“High” cell is the family-framing coefficient from a separate regression of insurance share on the framing indicator, TLU, and household size, with round and field-site fixed effects and standard errors (in parentheses) clustered at the session level. The Interaction column is the coefficient on High × Family Framing from a single interacted regression on that measure. The objective proxies are entered in place of the index, not as additional controls. \* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 25: Empowerment Moderator with Baseline-Proxy Interactions

| Specification                                     | Family Framing<br>(low empowerment) | High Emp. $\times$ Framing |
|---------------------------------------------------|-------------------------------------|----------------------------|
| Base (framing $\times$ high empowerment)          | 0.018***<br>(0.007)                 | −0.016<br>(0.010)          |
| + Locus of control (agency) $\times$ framing      | 0.019***<br>(0.007)                 | −0.016<br>(0.010)          |
| + Psychological distress (CES-D) $\times$ framing | 0.018***<br>(0.007)                 | −0.015<br>(0.010)          |
| + Realised livestock loss $\times$ framing        | 0.010<br>(0.029)                    | −0.015<br>(0.010)          |
| + Wealth ladder $\times$ framing                  | 0.018***<br>(0.007)                 | −0.015<br>(0.010)          |
| + Household head $\times$ framing                 | 0.018**<br>(0.007)                  | −0.017*<br>(0.010)         |
| + Livestock-decision power $\times$ framing       | 0.019**<br>(0.008)                  | −0.019<br>(0.012)          |

Dependent variable: share of budget allocated to insurance ( $N = 7,923$ ). Each row adds the named baseline proxy, both as a main effect and interacted with the framing treatment, to the base specification (family framing interacted with a high-empowerment indicator), with round and field-site fixed effects and standard errors clustered at the session level. “Family Framing” is the base coefficient, i.e. the effect among low-empowerment households; “High Emp.  $\times$  Framing” is the empowerment moderator. The moderator is stable in sign and magnitude across all specifications and is not significant at the 5% level in any of them. The low-empowerment base effect stays significant except where a nearly constant proxy (realized loss) inflates its standard error. Baseline proxies: locus-of-control (agency) index, psychological distress (CES-D), realized livestock loss in the prior year, a self-placed wealth ladder, whether the respondent is the household head, and subjective livestock-decision power. \* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .